Af Education Services

UACE MATHEMATICS ITEM BANK, PAPER 1

2026 Edition

FREE SAMPLE

2-in-1: Topical items (Chapters 1–15) + UNEB-standard items by construct

Over 1,500 scenario-based items · 900+ pages · Step-by-step responses

Get the full book: UGX 30,000

Calls/WhatsApp: **+256 759 336 820**

Email: afeducators.help@gmail.com

SAMPLE — Af Education Service

About This Sample

This sample gives you a taste of the **UACE Mathematics Item Bank, Paper 1**. It contains a few items taken directly from the full book:

- **Part One (Topical Items):** two items each from *Numerical Concepts*, *Differentiation*, *Vectors* and *Complex Numbers* – one from an exercise and one end-of-chapter item.
- **Part Two (Items by Construct):** one UNEB-standard item each from *Geometry*, *Algebra* and *Calculus*.

Every item is a real-life scenario from Uganda, and every task ends with a **decision**, just as UNEB now examines. Some items in this sample come with their full responses so that you can see how the book teaches. The rest are left for you to try; their responses are in the full book.

What the full book gives you

- All 15 chapters of Paper 1, exercise by exercise, with end-of-chapter items.
- 296 integrated UNEB-standard items across the three constructs.
- Over 1,500 items with step-by-step responses, so you can learn on your own.

How to use it: attempt each item first, then check the response. That is how the marks come.

Order your copy today: UGX 30,000

Calls/WhatsApp **+256 759 336 820** · afeducators.help@gmail.com

SAMPLE – Af Education Service

UACE PRINCIPAL MATHEMATICS PAPER 1

Constructs and the Topics They Cover

Every topic belongs to one of three constructs. The table shows which topics each construct covers, so you can see how the chapters of the book fit together.

No.	GEOMETRY	ALGEBRA	CALCULUS (ANALYSIS)
1	Coordinate geometry 1	Numerical concepts	Partial fractions
2	Trigonometry	Equations and inequalities	Differentiation 1
3	Vectors	Permutations and combinations	Differentiation 2
4	Coordinate geometry 2	Binomial theorem	Integration 1
5		Series	Integration 2
6		Complex numbers	Differential equations

Contents of the Full Book

Part One: Topical Items

Chapter 1: Numerical Concept
Chapter 2: Equations and Inequalities
Chapter 3: Coordinate Geometry
Chapter 4: Partial Fractions
Chapter 5: Trigonometry
Chapter 6: Differentiation 1
Chapter 7: Integration 1
Chapter 8: Permutations and Combinations
Chapter 9: Series
Chapter 10: Vectors
Chapter 11: Differentiation 2
Chapter 12: Integration 2
Chapter 13: Circles and Conic Sections
Chapter 14: Complex Numbers
Chapter 15: Differential Equations

Part Two: Items by Construct

Construct 1: Geometry (Items 1–80) · Construct 2: Algebra (Items 1–106) · Construct 3: Calculus (Items 1–110)

SAMPLE – Af Education Service

CHAPTER 1
NUMERICAL CONCEPT

Indices · Logarithms · Surds · End of Chapter Items

Exercise 1.2: Logarithms

Item 6.

Your brother works with a cybersecurity company testing a new encryption system in which access codes rest on logarithmic relationships. During the test, the security variable x satisfied $1 + 2\log_4 x = \log_x 4$, and the system settings accept only a whole-number value of x.

Two linked database variables, m and y, control the flow of information between servers. They satisfy $\log(y - m) = 0$ and $2\log y = \log(21 + m)$, and the network can be deployed only if both variables come out positive with m less than y; if not, the servers stay on the old network.

A senior analyst built a verification model on a parameter n with $\log_5 n = p$ and $\log_{75} n = q$. The analyst claims these imply $3^q = 5^{p-2q}$, and the verification model will be switched on only if the claim holds; otherwise, the old verification model stays on.

Tasks:
Help your brother to:

a) decide which value of the security variable the system settings accept.

b) determine whether the network can be deployed.

c) advise whether the verification model is switched on.

Exercise 1.4: End of Chapter Items

Item 2.

An agro-processing factory packs fruit juice for distribution across East Africa. While installing a new automated control system, the engineers set up several models to scale the equipment. A control variable $x = \dfrac{2\sqrt{24}}{\sqrt{2} + \sqrt{3}}$ sets one of the machine's adjustment ratios, $\dfrac{x + \sqrt{8}}{x - \sqrt{8}}$, and the adjustment dial only goes up to 25, so any larger ratio would need a new dial.

Two monitoring variables, x and y, give the pressure and temperature settings of the pasteurisation unit. They must satisfy $2\log_x y = 1$ and $xy = 64$, and the unit operates efficiently only if the pressure setting exceeds the temperature setting; otherwise, the unit will be recalibrated.

Three synchronisation variables, x, y and z, coordinate the filling, sealing and packaging machines through $2^x = 4^y = 8^z$ and $\dfrac{1}{2x} + \dfrac{1}{4y} + \dfrac{1}{8z} = \dfrac{22}{7}$. The automation specialist claims

SAMPLE - Af Education Service

these lead to $x = \frac{7}{16}$, $y = \frac{7}{32}$ and $z = \frac{7}{48}$, and the machines will be synchronised on these values only if the claim is true; if not, the specialist must recompute them.

Tasks:
Help the engineers to:

a) decide whether the dial can be set to the adjustment ratio.

b) determine whether the pasteurisation unit operates efficiently.

c) advise whether the machines can be synchronised on the specialist's values.

RESPONSES

Exercise 1.2, Item 6.

a) With $t = \log_4 x$, the equation becomes $1 + 2t = \dfrac{1}{t}$, so $2t^2 + t - 1 = 0$ and $t = \frac{1}{2}$ or $t = -1$, giving $x = 2$ or $x = \frac{1}{4}$. The system settings accept $x = 2$ but not $\frac{1}{4}$.

b) $y - m = 1$ and $y^2 = 21 + m = 20 + y$ give $y^2 - y - 20 = 0$, so $y = 5$ ($y = -4$ rejected) and $m = 4$. Both are positive and $m < y$, so the network can be deployed.

c) $n = 5^p = 75^q = 3^q \times 5^{2q}$, so $3^q = 5^{p-2q}$. The claim holds and the verification model is switched on.

Exercise 1.4, Item 2.

Full step-by-step response: in the full book.

SAMPLE – Af Education Service

CHAPTER 6
DIFFERENTIATION 1

Gradient · Tangents and Normals · Implicit functions · Turning points · Small changes · Rates of change · End of Chapter Items

---

Exercise 6.5: Nature of turning points and application of maxima and minima

Item 3.

Kato, your uncle, farms and runs a small bakery in the same compound. For a new livestock pen he has 100 m of metal railing for two adjacent sides of a rectangle, while two farm walls meeting at right angles form the other two. His animals need at least 2400 m^2 of grazing space; otherwise, he will buy more railing.

In the bakery he makes open cardboard boxes with square bases to pack cakes; each must hold 108 cm^3 and comes from a sheet of 120 cm^2 of cardboard; if one sheet is not enough, larger sheets will be bought.

Tasks:
Help Kato to:

a) decide whether the largest possible pen gives enough grazing space.

b) determine whether one sheet suffices for a box made with the least cardboard.

Exercise 6.10: End of Chapter Items

Item 5.

Your father sells eggs from his poultry farm. When he supplies q trays a week, his weekly revenue comes to $R = q\sqrt{1000 - q^2}$ thousand shillings. He plans to supply 25 trays a week, and he keeps that plan only if it lies within 2 trays of the supply that gives the greatest revenue; otherwise, he will adjust his supply to that level.

In his store, maize forms a conical heap whose height stays fixed while the base spreads, so that the volume grows by 2%. The rope pegged round the base of the heap snaps if the base circumference grows by more than 1.5%, in which case a longer rope will be used.

A fish pond on the farm has an edge that follows $x^2y + 3y^2 - 4x - 12 = 0$. A drainage pipe runs along the normal to the edge at $(0, 2)$, and it works only if it passes through the outlet at $(1, -1)$; otherwise, it will be relaid.

The access road to the farm has a speed hump whose profile follows $y = \dfrac{2}{1 + x^2}$, in decimetres. The district engineer approves a hump only if its single turning point forms a crest (a maximum) no higher than 1.5 dm; otherwise, the hump must be lowered.

Tasks:
Help your father to:

a) decide whether he should keep his plan of 25 trays a week.

b) determine whether the rope survives.

SAMPLE – Af Education Services

c) advise whether the drainage pipe works.

d) decide whether the engineer approves the hump.

RESPONSES

Exercise 6.5, Item 3.

a) With railing sides x and $100 - x$, the area $A = x(100 - x)$ is greatest when $\frac{dA}{dx} = 100 - 2x = 0$, at $x = 50$: a 50 m by 50 m pen of 2500 m^2, at least 2400 m^2. The pen gives enough grazing space.

b) With base side x and height h, $x^2h = 108$, so $A = x^2 + 4xh = x^2 + \frac{432}{x}$. Then $\frac{dA}{dx} = 2x - \frac{432}{x^2} = 0$ gives $x = 6$ cm, with height 3 cm, using $36 + 72 = 108$ cm^2, within 120 cm^2. One sheet suffices.

Exercise 6.10, Item 5.

Full step-by-step response: in the full book.

SAMPLE – Af Education Service

CHAPTER 10
VECTORS

Lines · Planes · Vector Geometry · End of Chapter Items

Exercise 10.2: Planes

Item 4.

Fibre-optic cable now runs along the main roads, and Miriam supervises the joints. One cable follows $\frac{x+1}{2} = \frac{y-3}{5} = \frac{z+1}{-1}$, in metres, and a junction box will go where it meets the control surface $x + y + z = 19$. Boxes cannot be buried deeper than 5 m, so the joint works only if it lies at $z \geq -5$; otherwise, the box will be moved.

A maintenance duct follows $\frac{x+2}{-1} = \frac{y-2}{2} = \frac{z-4}{1}$ and meets the surface $2x - y + 3z = 10$, where an access cover must sit exactly at ground level ($z = 0$); otherwise, the duct will be realigned. A cable marker at $(-2, 0, 6)$ must stay at least 1 m from that same surface; if not, it will be moved.

Tasks:
Help Miriam to:

a) decide whether the junction box can go where the cable meets the control surface.

b) determine whether the access cover and the cable marker meet their requirements.

Exercise 10.4: End of Chapter Items

Item 6.

Your cousin Ruth has just started as a site engineer and calls you over one Saturday to check her work. Anchor points A and B have position vectors $(3, -2, 5)$ and $(9, 1, -1)$, and a marker C divides AB in the ratio $5 : -3$. The marker can go in only if it lies within the site, where $x \leq 20$; otherwise, it will be relocated.

A foundation has corners $P(2, -1, 5)$, $Q(7, 1, -3)$ and $R(12, -2, 0)$, and her supervisor will approve it only if angle PQR comes out at exactly 90°, so that a fourth corner S completes a rectangle $PQRS$; otherwise, the corners will be reset.

A valve sits at $A(2, -1, 4)$ beside a pipe along $\mathbf{r} = (1, 0, 2) + \lambda(2, 1, 2)$, and the shortest connector from the valve to the pipe must come from a 2 m offcut in the store; otherwise, a new pipe will be bought.

A support rod OR, with $\overrightarrow{OR} = (2, 1, 2)$, must tilt at least 40° from the ground, the x-y plane; otherwise, it will be refitted.

Tasks:
Help Ruth to:

a) decide whether marker C can go in.

b) determine whether the supervisor will approve the foundation.

SAMPLE — Af Education Services

c) advise whether the 2 m offcut will do for the connector.

d) decide whether the support rod tilts enough.

RESPONSES

Exercise 10.2, Item 4.

a) A point on the cable is $(-1+2t,\ 3+5t,\ -1-t)$. In the surface: $1+6t=19$, so $t=3$ and the box goes at $(5, 18, -4)$. Since $z=-4 \geq -5$, the joint works.

b) A point on the duct is $(-2-t,\ 2+2t,\ 4+t)$. In the surface: $6-t=10$, so $t=-4$, giving $(2, -6, 0)$, exactly at ground level. The marker lies $\dfrac{|-4-0+18-10|}{\sqrt{14}} = \dfrac{4}{\sqrt{14}} \approx 1.07$ m from the surface, at least 1 m. Both requirements are met.

Exercise 10.4, Item 6.

Full step-by-step response: in the full book.

SAMPLE — Af Education Service

CHAPTER 14
COMPLEX NUMBERS

Complex Numbers · End of Chapter Items

Exercise 14.1: Complex Numbers

Item 6.

Your cousin, a sound engineer, builds speaker filters whose behaviour depends on the roots of polynomial equations. The bass filter follows $2z^3 - 9z^2 + 14z - 5 = 0$, and a supplier states that $2 + i$ forms one root. The filter works only if that statement holds and the equation's single real root lies between 0 and 1; otherwise, it will be redesigned.

The treble filter follows $z^4 - 5z^3 + 18z^2 - 17z + 13 = 0$, with $2 + 3i$ and $2 - 3i$ among its roots. It stays stable only if every root lies within 3 units of the origin on the Argand diagram; otherwise, a damping circuit will be added.

Tasks:
Help your cousin to:

a) decide whether the bass filter works.

b) determine whether the treble filter stays stable.

Exercise 14.2: End of Chapter Items

Item 1.

Your cousin, an electronics technician, builds sound systems for weddings and church events. His mixing software relies on two multiple-angle formulas that its manual derives from De Moivre's theorem: $\cos 6x = \cos^6 x - 15\cos^4 x \sin^2 x + 15\cos^2 x \sin^4 x - \sin^6 x$ and $\tan 3x = \dfrac{3\tan x - \tan^3 x}{1 - 3\tan^2 x}$. He trusts the software only if both formulas hold; otherwise, he will report the fault.

A filter in the mixer works with the values of $\left(1 + i\sqrt{3}\right)^{\frac{2}{3}}$. A colleague's notes list exactly two values, $1.2160 + 1.0204i$ and $-1.4917 + 0.5429i$, and the filter can use the notes only if that list is complete; otherwise, the notes will be completed.

Four speakers stand on a hall floor drawn as an Argand diagram, at the roots of $z^4 + 4 = 0$ found through De Moivre's theorem. The layout works only if the four speakers form a square of side 2 units; otherwise, the speakers will be moved.

A tuning point Z, with conjugate $\bar{Z}$, satisfies $Z\bar{Z} - 2Z + 2\bar{Z} = 5 - 4i$. The tuner works only if every possible value of Z has a positive real part; otherwise, it will be retuned.

Tasks:
Help your cousin to:

a) decide whether he can trust the software.

b) determine whether the filter can use the colleague's notes.

SAMPLE — Af Education Services

c) advise whether the speaker layout works.

d) decide whether the tuner works.

RESPONSES

Exercise 14.1, Item 6.

a) $(2+i)^2 = 3+4i$ and $(2+i)^3 = 2+11i$, so $2(2+11i) - 9(3+4i) + 14(2+i) - 5 = 0$ and $2+i$ is a root; so is $2-i$. Their product, $z^2 - 4z + 5$, divides the cubic: $2z^3 - 9z^2 + 14z - 5 = (z^2 - 4z + 5)(2z - 1)$, so the real root is $\frac{1}{2}$, between 0 and 1. The bass filter works.

b) $(z-2-3i)(z-2+3i) = z^2 - 4z + 13$, and $z^4 - 5z^3 + 18z^2 - 17z + 13 = (z^2 - 4z + 13)(z^2 - z + 1)$, so the other roots are $\frac{1 \pm i\sqrt{3}}{2}$, with modulus 1. But $|2 \pm 3i| = \sqrt{13} \approx 3.61$, beyond 3. The treble filter does not stay stable, so a damping circuit will be added.

Exercise 14.2, Item 1.

Full step-by-step response: in the full book.

SAMPLE — Af Education Service

PART TWO

ITEMS BY CONSTRUCT

Geometry · Algebra · Calculus (Analysis)

UNEB-standard items, each drawing on several topics within one construct. The full book has 296 such items with responses.

SAMPLE – Af Education Service

CONSTRUCT 1
GEOMETRY

Coordinate geometry 1 · Trigonometry · Vectors · Coordinate geometry 2

Item 51.

Traders at the Busia border market have asked the town council to reorganise the market grounds. On the council's site map, with one unit representing 100 m, a feeder road runs along $5x - 12y + 18 = 0$ and a second road along $12x + 5y - 92 = 0$. The two roads meet at the main junction J, where the council will build a market office. A borehole already stands at $B(13, -3)$, and to keep dust from trucks out of the water, it must be at least 400 m from the second road. If it stands any closer, the council will sink a new borehole elsewhere.

Goods are wheeled up a loading ramp into the main shed. The ramp's angle of inclination θ satisfies $6\sin^2\theta + \sin\theta\cos\theta - 2\cos^2\theta = 0$, where θ is acute. Wheelbarrow pushers can only manage ramps inclined at not more than 25°; a steeper ramp must be redesigned.

The shed is roofed with a flat canopy resting on three posts with tops at $P(2, 1, 4)$, $Q(5, 3, 4)$ and $R(3, -1, 6)$, in metres. At night a floodlight lights the canopy with a beam along the direction $\mathbf{i} - 2\mathbf{j} - 3\mathbf{k}$. The electrician wants the beam to strike the canopy nearly head-on, at an angle of at least 80° to the canopy surface, so that the whole roof is lit evenly. A beam striking at a smaller angle means the floodlight will be raised higher above the canopy.

Tasks:

a) Advise the council on where the market office will stand and whether a new borehole must be sunk.

b) Decide whether the loading ramp must be redesigned.

c) Advise the electrician on whether the floodlight is well positioned.

RESPONSES

Item 51.

a) Solving $5x - 12y = -18$ and $12x + 5y = 92$ gives $J(6, 4)$. The distance of B from the second road is $\dfrac{|156 - 15 - 92|}{13} = \dfrac{49}{13} = 3.77$ units $= 377$ m. The office stands at $(6, 4)$; 377 m is less than 400 m, so a new borehole must be sunk.

b) $6\tan^2\theta + \tan\theta - 2 = 0$, so $(3\tan\theta + 2)(2\tan\theta - 1) = 0$; for acute θ, $\tan\theta = \frac{1}{2}$ and $\theta = 26.6^\circ$. Since 26.6° exceeds 25°, the ramp must be redesigned.

c) $\overrightarrow{PQ} \times \overrightarrow{PR} = (4, -6, -8)$, so the canopy is $2x - 3y - 4z = -15$. Then $\sin\phi = \dfrac{|2 + 6 + 12|}{\sqrt{14}\sqrt{29}} = 0.993$, so $\phi = 83.0^\circ$. This is at least 80°, so the floodlight is well positioned and stays where it is.

SAMPLE – Af Education Service

CONSTRUCT 2
ALGEBRA

Numerical concepts · Equations and inequalities · Permutations and combinations · Binomial theorem · Series · Complex numbers

---

Item 80.

At a fish landing site in Nakasongola, traders sell tilapia, mukene and Nile perch.

A hotel bought 1 kg of tilapia, 2 kg of mukene and 1 kg of Nile perch for Ugx 26,000; 2 kg of tilapia, 1 kg of mukene and 1 kg of Nile perch for Ugx 31,000; and 3 kg of tilapia, 1 kg of mukene and 2 kg of Nile perch for Ugx 51,000. Its next order is 5 kg of tilapia, 10 kg of mukene and 4 kg of Nile perch, and the hotel has budgeted Ugx 120,000 for it. Should the order cost more, the quantity of Nile perch will be cut.

The weekly catch falls by 6% each week from 800 kg in the first week. The traders' cooperative needs at least 6000 kg over the first 10 weeks to meet its contracts. A smaller catch would force it to buy fish from Kalangala.

A patrol boat moves so that it is always equally far from the landing site at $3i$ and the fuel depot at -2 on an Argand diagram. The marine officer claims that a boat seen at $-1 + 1.5i$ is on this patrol path. A boat off the path will be stopped and checked.

The fish in a cage grow in number according to $N = 1200(1.15)^t$, where t is the time in months. The farmer will harvest when there are 10,000 fish, and he has promised a buyer a harvest within 15 months. If he cannot keep it, he will add more fingerlings to the cage.

Tasks:

a) Decide whether the hotel's budget is enough.

b) Decide whether the catch will meet the contracts.

c) Verify the marine officer's claim.

d) Decide whether the farmer will keep his promise.

Full step-by-step response: in the full book.

SAMPLE – Af Education Service

CONSTRUCT 3
CALCULUS (ANALYSIS)

Partial fractions · Differentiation 1 and 2 · Integration 1 and 2 · Differential equations

Item 11.

Mukasa is planning improvements on his dairy farm near a straight river bank. The raised boundary of one section of the farm is modelled on a ground plan by the curve $y = x^3 - 6x^2 + 9x + 5$. Mukasa wants to sketch this curve and identify its turning points, because the high and low points affect how rainwater flows towards the river and where animal shelters can safely be built. He will build the animal shelter at the local maximum only if it is at least 3 units higher than the local minimum, so that rainwater drains away from the shelter.

He also wishes to enclose a rectangular grazing section of area 1250 m^2, with one side lying along the straight river bank so that no barbed wire is needed on that side. Since barbed wire is expensive and trial fencing would waste materials, he wants to use the least possible length of barbed wire. For cost effectiveness, this length must not exceed 120 m and the total distance around the enclosed section must not exceed 200 m.

Within the same farm layout, two curved drainage boundaries are represented on the ground plan by the curves $y = x^2 + 1$ and $y = -x^2 + 4x + 1$. Mukasa plans to plant special grass in the region enclosed between the two curves to reduce soil erosion near the river channel and improve animal feeding during dry periods. He will buy the grass seed if the area of the region is more than 2 m^2.

After milking, fresh milk is heated for sterilisation and then transferred into a cooling chamber whose surrounding temperature remains constant at 20°C. The rate at which the milk cools is proportional to the difference between its temperature and the surrounding air temperature. Records show that the milk cools from 100°C to 70°C in 12 minutes. The milk can only be packed safely when its temperature has fallen to 40°C, and the farm loses money whenever cooling takes longer than 40 minutes, because workers and packaging machines remain idle.

Tasks: Before committing resources to the farm improvements,

a) analyse the land profile and decide whether the shelter should be built at the high point.

b) decide whether the fencing and grass-planting plans support efficient use of the available land.

c) advise whether packaging can begin within the required time without causing avoidable losses.

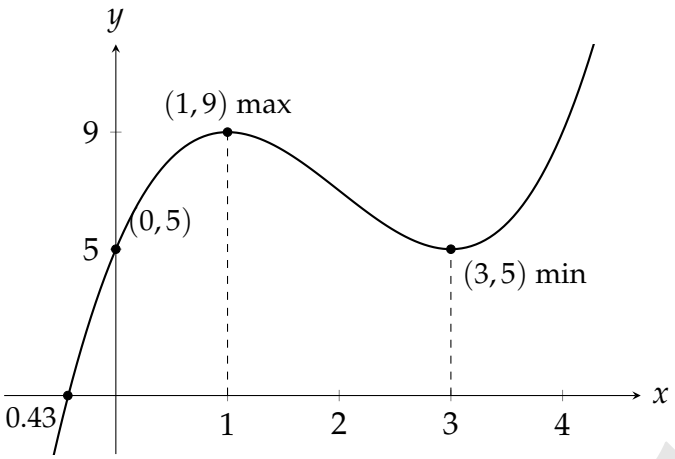
RESPONSES

Item 11.

a) $\dfrac{dy}{dx} = 3x^2 - 12x + 9 = 3(x-1)(x-3)$; $\dfrac{d^2y}{dx^2} = 6x - 12$ gives a maximum at $(1, 9)$ and a minimum at $(3, 5)$. The curve meets the y-axis at $(0, 5)$ and crosses the x-axis once,

SAMPLE – Af Education Service

near $x = -0.43$. The maximum is $9 - 5 = 4$ units above the minimum, at least 3, so the shelter should be built at the high point.



Sketch of $y = x^3 - 6x^2 + 9x + 5$

b) With x m for each side at right angles to the river and y m along it, $xy = 1250$ and the wire is $L = 2x + \dfrac{1250}{x}$. Then $\dfrac{dL}{dx} = 2 - \dfrac{1250}{x^2} = 0$ gives $x = 25$, $y = 50$ and $L = 100$ m; the distance around the section is 150 m. The curves meet where $2x^2 = 4x$, at $x = 0$ and $x = 2$, so the grass region is $\displaystyle\int_0^2 (4x - 2x^2)\,dx = \frac{8}{3} = 2.67$ m^2. The wire (100 m) and the distance around (150 m) are within their limits, so the fencing is cost-effective; 2.67 m^2 is more than 2 m^2, so the grass seed should be bought.

c) $\dfrac{dT}{dt} = -k(T - 20)$ gives $T = 20 + 80e^{-kt}$, and $70 = 20 + 80e^{-12k}$ gives $e^{-12k} = \frac{5}{8}$. So $T = 20 + 80\left(\frac{5}{8}\right)^{t/12}$. For $T = 40$: $\left(\frac{5}{8}\right)^{t/12} = \frac{1}{4}$, so $t = \dfrac{12 \ln 4}{\ln 1.6} = 35.4$ minutes. This is within 40 minutes, so packaging can begin in time and there are no avoidable losses.

SAMPLE – Af Education Service

You have seen 11 items.
The full book has over 1,500.

- All 15 Paper 1 chapters, exercise by exercise
- 296 UNEB-standard items by construct
- Step-by-step responses to every item
- Real Ugandan scenarios with decision-making tasks
- 900+ pages, 2026 Edition

UACE Mathematics Item Bank, Paper 1

UGX 30,000

Calls/WhatsApp: **+256 759 336 820**
afeducators.help@gmail.com

Af Education Services

Attempt every item before checking the response. Let's get that "A" together.

SAMPLE — Af Education Service